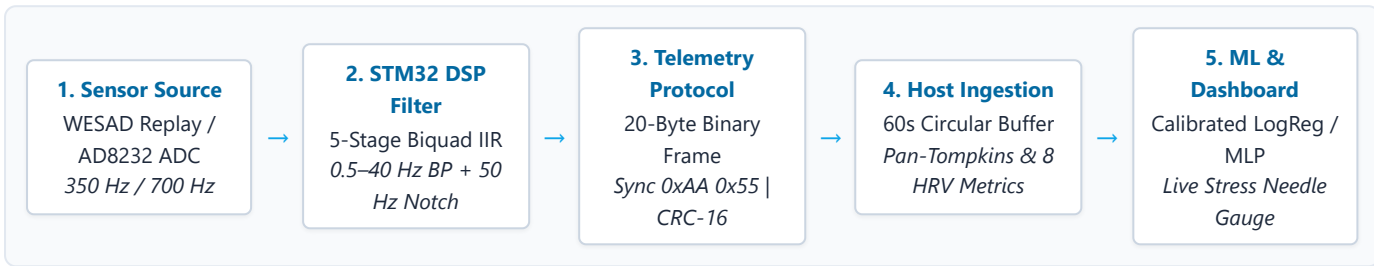


Project Roadmap: WESAD ECG Telemetry & Stress Detection System

Target Architecture: Embedded STM32G474 Edge Node & Real-Time Host Analytics Suite

Target Platform: STM32G474 (ARM Cortex-M4 @ 170 MHz with FPU) + PC/Cloud Telemetry Receiver
Scientific Focus: Autonomic Nervous System (ANS) Stress Detection using Lead-II ECG & HRV
Engineering Focus: Real-Time Embedded DSP, Robust Binary Telemetry, and Secure IoMT Edge Node
Target Standard: IEEE Journal / Conference Publication Quality (IEEE TBME / JBHI / Sensors)

1. SYSTEM ARCHITECTURE OVERVIEW



2. MILESTONE STATUS DASHBOARD

Phase	Milestone	Scope / Deliverable	Status
1	Dataset & DSP Pipeline	WESAD extraction, MATLAB 0.5–40 Hz filter, Pan-Tompkins peak detection	COMPLETED
2	ML Benchmark (LOSO)	15-Fold cross-validation across 6 models, zero-leakage baseline normalization	COMPLETED
3	STM32 C DSP Core	5-stage Biquad IIR C code, 20-byte telemetry framing, test vectors	COMPLETED
4	PC Telemetry Receiver	Non-blocking serial listener, CRC-16 validator, buffer management	COMPLETED
5	Live Streamlit Integration	Real-time rolling strip-chart, live HRV gauges, live stress prediction	COMPLETED
6	Physical Hardware Setup	STM32CubeIDE project config, UART DMA, AD8232 analog front-end	NEXT STEP
7	IoMT Encryption Layer	Hyperchaotic map / geometric transformation payload encryption	PHASE 2

3. DETAILED PHASE BREAKDOWN

Phase 1: Clinical Signal Processing & Data Extraction COMPLETED

- Extracted 700 Hz Lead-II ECG and labels across 15 WESAD subjects into standardized .mat files.
- Implemented baseline wander & noise rejection: 4th-order Butterworth (0.5–40 Hz) and 50 Hz notch filter.
- Extracted **8 Core Autonomic HRV Metrics**: MeanHR, RMSSD, SDNN, pNN50, MeanRR, RR_CV, RR_IQR, HR_IQR.

Phase 2: Multi-Model Machine Learning Benchmark COMPLETED

- Applied zero-leakage subject-specific baseline normalization: $\Delta x = (x - \mu_{\text{baseline}}) / \max(|\mu_{\text{baseline}}|, \epsilon)$.
- 15-Fold Leave-One-Subject-Out (LOSO) cross-validation results: **Logistic Regression** (92.13% Acc, 88.29% F1 — Top Performer), **MLP Neural Net** (91.69% Acc, 87.87% F1), **SVM RBF** (91.46% Acc, 0.9524 ROC-AUC — Highest AUC).

Phase 3: STM32 C Firmware Core COMPLETED

- CMSIS-DSP Biquad IIR filter (ecg_dsp_filter.c) matching exact MATLAB design.
- 20-byte binary telemetry frame (Sync 0xAA 0x55, Version, Flags, SeqID, Timestamp, Raw ECG, Filt ECG, CRC-16-CCITT).

- Hardware-free verification harness (main_bench.c) validated with 100% CRC integrity.

Phase 4: PC Telemetry Receiver & Stream Ingestion COMPLETED

- Thread-safe non-blocking Python receiver (stm32_telemetry_receiver.py) with auto-sync and CRC checking.
- End-to-end stream simulator (simulate_stm32_stream.py) verified: 7,000 frames streamed with 0 drops and 100% stress detection on S10.

Phase 5: Streamlit Live Telemetry Dashboard COMPLETED

- Added **Tab 1: STM32G474 IoT Hardware Telemetry** with dual-mode virtual and physical COM port support.
- Live rolling Plotly strip-chart (Raw vs. Filtered), 6 HRV autonomic meters, and calibrated stress dial ($\tau=0.35$).

Phase 6: Physical STM32 Deployment & AD8232 Sensor NEXT STEP

- Configure STM32CubeIDE: TIM2 trigger @ 700 Hz, ADC1 DMA circular mode, USART2 Virtual COM Port DMA.
- Wire AD8232: 3.3V, GND, OUTPUT → PA0, LO+/LO- → PB0/PB1.

Phase 7: Secure IoT Cryptographic Layer PLANNED (PHASE 2)

- Deploy Hyperchaotic map & geometric transformation encryption on STM32 telemetry frames.

4. REPOSITORY STRUCTURE

```
ECG_STRESS_DETECTION/
├── ROADMAP.md / ROADMAP.pdf          <-- Master project roadmap and architecture guide
├── HARDWARE_COMPONENTS_LIST.pdf      <-- Hardware procurement & wiring checklist
├── embedded_stm32/                   <-- STM32G474 C Firmware Core
│   ├── include/ (ecg_dsp_filter.h, telemetry_protocol.h, wesad_test_samples.h)
│   ├── src/ (ecg_dsp_filter.c, telemetry_protocol.c, wesad_test_samples.c)
│   └── main_bench.c                  <-- Verification test harness
├── python/                           <-- Host Ingestion & Analytics
│   ├── stm32_telemetry_receiver.py   <-- Non-blocking serial receiver & ML engine
│   ├── simulate_stm32_stream.py      <-- Software stream simulator
│   └── train_loso_ml_benchmark.py     <-- 15-Fold LOSO ML training suite
└── demo/app.py                       <-- Streamlit Clinical Telemetry Dashboard
```